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Guideline/Protocol Title	Guidelines for the Management of Pediatric Patients with Suspected Community-Acquired Bacterial Meningitis
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Committee Review	Antimicrobial Stewardship Subcommittee Pharmacy and Therapeutics Committee
Target Population	Pediatric patients with suspected bacterial meningitis
Overview	This guideline provides evidence-based management of community-acquired bacterial meningitis in pediatrics
Effective Date	8/2024
Revised Date	N/A
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Schedule for Periodic Review	Every 3 years
Implementation Strategy	Updated guideline will be shared on CareWeb and the UK Antimicrobial Stewardship website
Education Strategy	N/A
Primary Outcome(s)	N/A
Outcome Assessment Plan	N/A
Information Technology Needs	Access to CareWeb, UK Antimicrobial Stewardship website

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PURPOSE OF GUIDELINE: To provide evidence-based guidance for the management of pediatric patients with suspected community-acquired bacterial meningitis at the University of Kentucky HealthCare.

EXCLUSION: This guideline is NOT intended for the management of patients with confirmed or suspected malignancy.

MENINGITIS DEFINITION: Infection resulting in acute inflammation of the brain and spinal cord meninges.

SYMPTOMS OF BACTERIAL MENINGITIS: Common symptoms of bacterial meningitis include headache, phonophobia, photophobia, nausea/vomiting, neck stiffness, fever, encephalopathy, seizures, positive Babinski sign, or positive Kernig sign. Signs and symptoms of meningitis in young children and infants are often less reliable and nonspecific and include fever, lethargy, irritability, difficulty feeding, seizures, poor tone, and vomiting.

DIAGNOSIS AND INITIAL WORKUP OF MENINGITIS: Obtain STAT **blood culture** and STAT **lumbar puncture*** (LP) prior to antimicrobial administration but *never delay administration of antibiotics at time of or prior to LP if difficulty obtaining blood culture(s) or performing LP*.

- Order CSF studies*
  - Meningitis-encephalitis (ME) panel
  - Cell counts
  - Protein and glucose
  - Gram stain & culture
  - If clinical suspicion for neurosyphilis, obtain Venereal Disease Research Laboratory (VDRL) on CSF in addition to serum syphilis antibody with reflex to RPR
  - May consider AFB/fungal cultures and stains of CSF in certain clinical scenarios

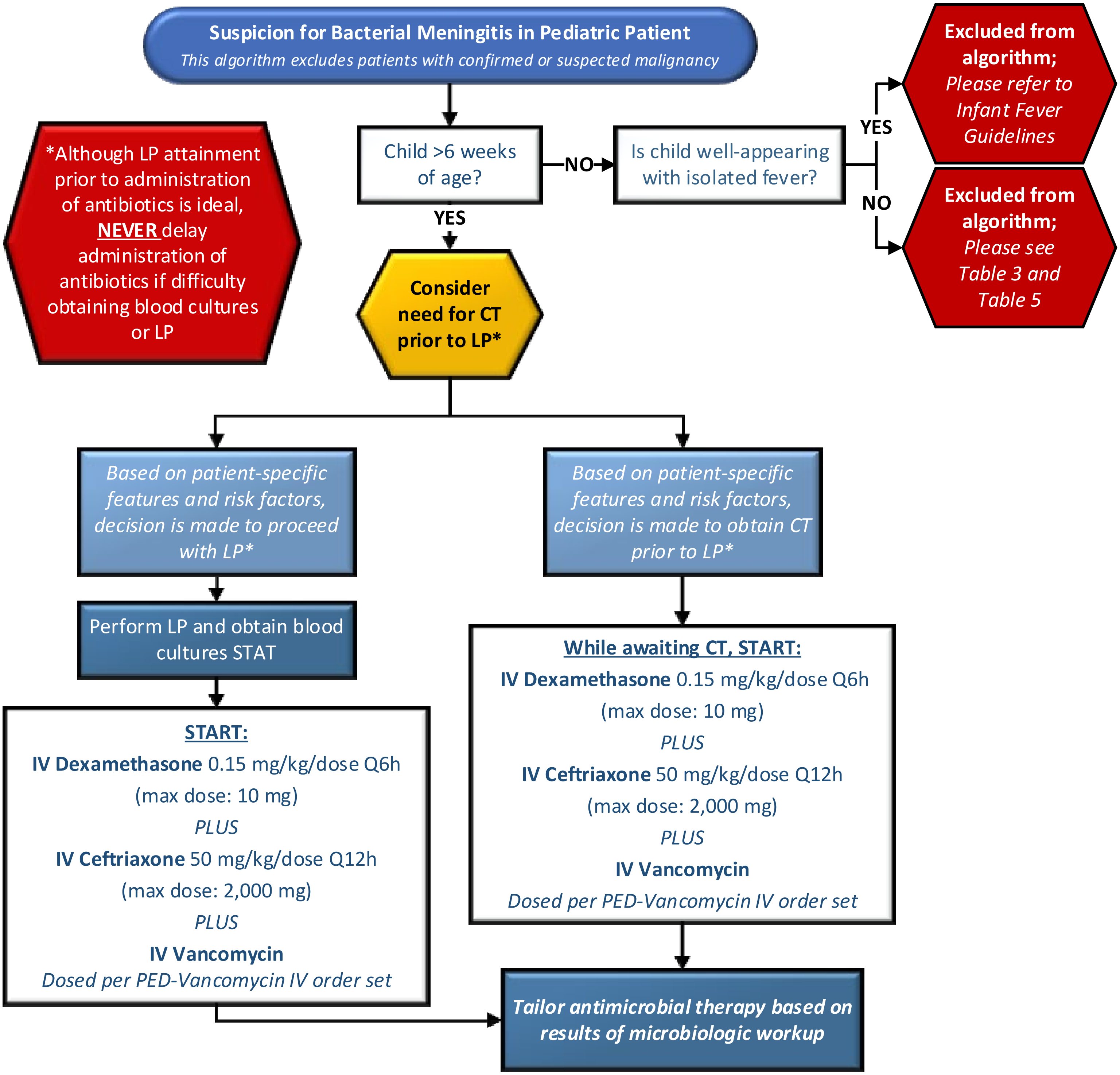
***In the setting of limited CSF volume prioritize:**
1. Meningitis-encephalitis (ME) panel
2. Cell counts
3. CSF protein/glucose

- Administer appropriate antibiotics within **30 minutes** of establishing clinical suspicion for meningitis. Do not delay administration of antibiotics if unable to obtain blood cultures or perform the LP.
- Dexamethasone should be administered in children >6 weeks of age when meningitis is suspected and ideally should be administered **PRIOR TO** or with the **FIRST DOSE** of antibiotics to optimize efficacy. Dexamethasone should be continued for a total of 4 days in children with *S. pneumoniae* or *H. influenzae* meningitis and discontinued in patients with meningitis secondary to other organisms. Of note, it is reasonable to administer dexamethasone after the first dose of antibiotics have been given. See Appendix B for additional details.

CONSIDERATIONS FOR IMAGING PRIOR TO LUMBAR PUNCTURE:

- The risk of brain herniation due to a LP in the sitting of a space occupying lesion or increased intracranial pressure (ICP) is mentioned in literature. However, there is overall poor-quality evidence supporting a causal relationship between a LP and herniation thus the concern is likely overstated. Providers should consider imaging prior to LP in cases where a diagnosis other than meningitis is strongly suspected, or in patients with severe impairment of consciousness or multiple of the factors described in the next bullet point. Additionally, if imaging is performed discussion of the findings with a subspecialist prior to performing a LP is reasonable.
- There are select clinical features that may predict CT findings of space occupying lesion or signs of increased ICP. *Considerations for ordering head imaging prior to LP include:*
  - Immunocompromised patients (e.g., HIV, AIDS, on immunosuppressive therapy, or post-transplantation)
  - New onset seizures (within 1 week of presentation)
  - History of CNS disease (mass, lesion, stroke, or focal infection)
  - Papilledema
  - Altered level of consciousness (especially GCS <10)
  - Focal neurological deficit

FIGURE 1: Algorithm for Initial Management of Patients with Concern for Acute Bacterial Meningitis



*: There are select clinical features that may predict CT findings of space occupying lesion or signs of increased intracranial pressure (ICP) and it may be prudent for providers to consider imaging in cases of severe impairment in consciousness or multiple of these factors and discussion with subspecialist regarding imaging findings prior to performing LP. Please see Appendix A for consensus guideline definitions of patients at high risk for abnormal CT.

TABLE 1: Interpreting CSF Studies

CSF Study	Normal	Bacterial Meningitis	Fungal Meningitis	Viral Meningitis
Appearance	Clear	Turbid	Clear or Turbid	Clear
Opening Pressure	$\leq$20 cm H_2O	High	High	Normal to High
Glucose	2/3 of serum	Low (<40 mg/dL)	Low (<40 mg/dL)	Normal
Protein	<50 mg/dL	High (100-500 mg/dL)	High (>45 mg/dL)	Normal to High (generally <200 mg/dL)
Nucleated Cell Count (WBC)	<5 cells/uL	High, predominantly neutrophilic	High, neutrophilic and lymphocytic	High, predominantly lymphocytic

TABLE 2: Meningitis-Encephalitis Panel

Meningitis-Encephalitis Panel		
Bacteria	**Virus**	**Fungal**
Escherichia coli K1 *Haemophilus influenzae* *Listeria monocytogenes* *Neisseria meningitidis* *Streptococcus agalactiae* *Streptococcus pneumoniae*	*Cytomegalovirus* (CMV) *Enterovirus* (EV) *Herpes simplex virus 1* (HSV-1) *Herpes simplex virus 2* (HSV-2) *Human herpes virus 6* (HHV-6) *Human parechovirus* (HPeV) *Varicella zoster virus* (VZV)	*Cryptococcus neoformans/gattii*

The BioFire® Meningitis/Encephalitis (ME) Panel uses PCR technology to detect the following organisms:

TABLE 3: Empiric Therapy for Meningitis

Patient Population	Common Pathogens	Empiric Regimen
	Community-Acquired	
Age ≤4 weeks *Please see Table 5 for guidance on neonatal dosing of antibiotics*	*S. agalactiae* (GBS), *E. coli,* Herpes simplex virus (HSV), *L. monocytogenes*^	**IV Ampicillin*** *PLUS* **IV Cefotaxime*** *OR* I**V Ceftriaxone**[#] *PLUS* **IV Acyclovir** 20 mg/kg/dose Q8h **Refer to* Table 5 *for neonatal dosing of cefotaxime and ampicillin*
>4 weeks to 6 weeks	*S. agalactiae* (GBS), *E. coli,* +/- HSVɸ	**IV Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg) *Recommend complete HSV evaluation and addition of IV acyclovir 20 mg/kg/dose Q8h in the setting of maternal or infant risk factors*[ɸ]
Age >6 weeks to 23 months	*S. pneumoniae, N. meningitidis, H. influenzae, S. agalactiae, E. coli*	**IV Vancomycin** *dosed per PED-Vancomycin IV order set* *PLUS* **IV Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg) *PLUS* **IV Dexamethasone** 0.15 mg/kg/dose Q6h** (max dose: 10 mg)
Age 2 to 18 years	*N. meningitidis, S. pneumoniae*	
Suspicion for HSV[ɸ]	Herpes simplex virus	*≤3 months*: **IV Acyclovir** 20 mg/kg/dose Q8h *>3 months*: **IV Acyclovir** 10 mg/kg/dose Q8h
	Trauma/Surgical History	
Post-Neurosurgery[¥] **Post-Trauma**[¥]	*S. aureus;* coagulase-negative *Staphylococci;* aerobic Gram-negative bacilli (including *P. aeruginosa*)	**IV Vancomycin** *dosed per PED-Vancomycin IV order set* *PLUS* **IV Cefepime** 50 mg/kg/dose Q8h (max dose: 2,000 mg)
Concern for CSF shunt infection[¥]	*S. aureus,* coagulase-negative staphylococci*;* aerobic Gram-negative bacilli (including *P. aeruginosa*), *Cutibacterium acnes* (previously *Propionibacterium acnes)* **NOTE:** Please ensure the microbiology lab is aware of specimens obtained from shunts or drains as cultures must be held for at least 10 days to identify slow-growing organisms including *C. acnes*	**IV Vancomycin** *dosed per PED-Vancomycin IV order set* *PLUS* **IV Cefepime** 50 mg/kg/dose Q8h (max dose: 2,000 mg)

#Ceftriaxone may be used in neonates with postmenstrual age (PMA) ≥42 weeks who are not receiving calcium-containing fluids and who do not have pre-existing hyperbilirubinemia; ****Dexamethasone:** Ideally administer dexamethasone prior to or with first dose of antibiotics for maximal efficacy; Dexamethasone is **NOT** recommended in post-neurosurgical or post-traumatic meningitis cases. IV to PO conversion may be considered following clinical improvement if able to tolerate PO; **^*Listeria*** coverage should be empirically initiated in patients who are <4 weeks of age, pregnant, or immunosuppressed. In the setting of a life-threatening beta-lactam allergy, see Table 4; **ɸHerpes simplex virus (HSV):** See Appendix C for HSV risk factors in children >4 weeks of age; **¥:** When considering risk of infection in patients with history of neurosurgical manipulation (craniotomy, craniectomy, spinal manipulation, etc.), traumatic head injury, or CSF shunt placement, it is important to consider time from injury/surgical manipulation and history of shunt revision.

Table 4: Definitive Therapy for Meningitis

Once a pathogen is isolated, tailor antibiotic regimen to identified organism. Choice of specific antimicrobial agent must be guided by in-vitro susceptibility test results. In the setting of *Streptococcus pneumoniae* and *Neisseria meningitidis*, MIC breakpoints for meningitis must be carefully applied to determine the choice of therapy:

Microorganism	CSF MIC#	Preferred Therapy	Alternative Therapy^
Streptococcus pneumoniae	Penicillin MIC ≤0.06 ug/mL	**Aqueous Penicillin G** 66,667 units/kg/dose Q4h (max dose: 4 million units)	**Ampicillin** 66.7 mg/kg/dose Q4h (max dose: 2,000 mg) *OR* **Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg)
	Penicillin MIC >0.06 ug/mL & Ceftriaxone MIC ≤0.5 ug/mL	**Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg)	**Linezolid** 10 mg/kg/dose Q8-12h# (max dose: 600 mg) *#Q8h if <12 years, Q12h if ≥12 years*
	Penicillin MIC >0.06 ug/mL & Ceftriaxone MIC ≥1 ug/mL	**Vancomycin** *dosed per PED-Vancomycin IV order set* **PLUS** **Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg)	**Linezolid** 10 mg/kg/dose Q8-12h# (max dose: 600 mg) *#Q8h if <12 years, Q12h if ≥12 years*
Neisseria meningitidis	Penicillin MIC ≤0.06 ug/mL	**Aqueous Penicillin G** 66,667 units/kg/dose Q4h (maximum: 4 million units)	**Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg)
	Penicillin MIC >0.06 ug/mL & Ceftriaxone MIC ≤0.12 ug/mL	**Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg)	**Meropenem*** 50 mg/kg/dose Q8h (max dose: 2,000 mg)

***Dosing provided for children >1 month of age. Refer to Table 5 for guidance on dosing in neonates;* #:** Ensure that appropriate CNS breakpoints applied, as outlined in table, for treatment of meningitis; **^ID Consult** is strongly recommended for patient cases in which alternative therapy is considered; ***Meropenem MIC for *N. meningitidis* must be ≤0.25 ug/mL to be considered susceptible.

Microorganism	Preferred Therapy	Alternative Therapy^
Listeria monocytogenes	**IV Ampicillin** 66.7 mg/kg/dose Q4h (max dose: 2,000 mg)	**IV Aqueous Penicillin G** 66,667 units/kg/dose Q4h (max dose: 4 million units) *OR* **IV/PO TMP/SMX** 5 mg/kg/dose IV/PO Q8h (max dose: 320 mg) *based on trimethoprim component* *OR* **IV Meropenem*** 50 mg/kg/dose Q8h (max dose: 2,000 mg)
Salmonella	**IV Ceftriaxone** 50 mg/kg/dose Q12h (max dose: 2,000 mg)	**IV/PO TMP/SMX** 5 mg/kg/dose IV/PO Q8h (max dose: 320 mg) *based on trimethoprim component* *OR* **IV Meropenem*** 50 mg/kg/dose Q8h (max dose: 2,000 mg)
***Treponema pallidum* (Syphilis)**	**IV Aqueous Penicillin G** 66,667 units/kg/dose Q4h‡ (max dose: 4 million units)	**CONSULT ID**

***Dosing provided for children >1 month of age. Refer to Table 5 for guidance on dosing in neonates;* Abbreviations**: TMP/SMX: trimethoprim/sulfamethoxazole; **^ID Consult** is strongly recommended for patient cases in which alternative therapy is considered; ***Meropenem** is an alternative agent that can be employed in the setting of an inability to use either standard therapy or TMP/SMX;

‡Syphilis: Experts recommend administration of penicillin G benzathine 50,000 units/kg/dose (max dose: 2.4 million units) once weekly for 1-3 weeks following completion of parenteral aqueous penicillin G for treatment of neurosyphilis.

Table 5: Antibiotic Dosing in Neonates

Ampicillin	
Postnatal age	**Dose**
≤7 days	Ampicillin 100 mg/kg/dose Q8h
8 to 28 days	Ampicillin 75 mg/kg/dose Q6h
Aqueous Penicillin G	
Postnatal age	**Dose**
≤7 days	Aqueous Penicillin G 150,000 units/kg/dose every 8 hours
8 to 42 days	Aqueous Penicillin G 125,000 units/kg/dose every 6 hours
Cefotaxime	
Postnatal age	**Dose**
≤7 days	Cefotaxime 50 mg/kg/dose every 8 hours
8 to 42 days	Cefotaxime 50 mg/kg/dose every 6 hours

Table 6: Duration of Therapy

Microorganism	Duration of Therapy (days)
Neisseria meningitidis	7
Haemophilus influenzae	7
Streptococcus pneumoniae	10-14
Staphylococcus spp.	10-14
Cutibacterium (Propionibacterium) acnes	10-14
Streptococcus agalactiae	14-21
Aerobic Gram-negative bacilli*	14-21*
Listeria monocytogenes	≥21
Salmonella	≥28
Herpes Simplex Virus (HSV)	21
Neurosyphilis	10-14

Examples of aerobic Gram-negative bacilli include the Enterobacterales order (i.e. *E. coli, Klebsiella spp., Enterobacter spp., Citrobacter spp.*), *Pseudomonas aeruginosa,* and *Acinetobacter spp.* In neonates, duration of therapy for aerobic gram-negative meningitis should be 2 weeks from CSF culture sterility or ≥3 weeks total, whichever is longer.

- If suspecting **culture-negative meningitis**, continue empiric antibiotic therapy for 7-21 days depending on the likelihood of pathogen based off on the patient's age and risk factors (see Table 3: Empiric Therapy)
- **ID CONSULT** is recommended in all cases of bacterial meningitis.
- **Interpretation of delayed LP** results is often challenging and ID consultation is recommended in this scenario. Available data suggests that the sensitivity of the HSV CSF PCR is nearly 100% if obtained within 7 days of acyclovir initiation.[5]
- **Repeat LP** is only recommended in patients who have not responded clinically after 48 hours of appropriate antimicrobial therapy. Repeat LP may be considered for patients receiving concomitant dexamethasone and for ceftriaxone-resistant *S. pneumoniae* meningitis. Otherwise, repeat CSF analysis to document sterilization is not routinely indicated.
- **In patients with neurosyphilis,** some experts recommend administration of penicillin G benzathine 50,000 units/kg/dose (max dose: 2.4 million units) once weekly for 1-3 weeks following completion of parenteral aqueous penicillin G when staging (e.g., early vs. latent) unknown.

APPENDICES:

Appendix A: Consensus Guideline Definitions of High Risk for Abnormal Head CT

Infectious Diseases Society of America (IDSA)	European Society of Clinical Microbiology and Infectious Diseases (ECMID)
Glasgow Coma Scale (GCS) <15	GCS <10
Focal neurological deficit	Focal neurological deficit
Severely immunocompromised (transplant recipient or HIV infection)	Severely immunocompromised (transplant recipient or HIV infection)
History of new-onset seizure within 1 week	History of new-onset seizure within 1 week
History of intracranial lesion	
Papilledema	

Table adapted from Regina J et al, *J Hosp Med*. 2023; 18: 534-537.

Appendix B: Role of corticosteroids

- Dexamethasone is recommended, in combination with antibiotics, in pediatric patients >6 weeks of age with suspected bacterial meningitis and should be continued for a total of 4 days in children with *S. pneumoniae* or *H. influenzae* meningitis. Dexamethasone should be discontinued in patients with meningitis secondary to other organisms.
- **Timing of administration**: As the majority of randomized controlled trials administered dexamethasone with the first dose of antibiotics, it is recommended that corticosteroids be administered prior to or with the first dose of antibiotics.
    - *There is a paucity of high-quality data supporting a specific timeframe in which dexamethasone should be given. It is reasonable to administer steroids after the first dose of antibiotics. This recommendation is based on expert opinion and accounts for possible benefit with likely low likelihood of harm.*
- **Rationale:** Animal studies have demonstrated that the negative outcome(s) of bacterial meningitis is related to severity of inflammation in the subarachnoid space.[7] A Cochrane meta-analysis, which included 25 randomized controlled trials and evaluated 4,121 pediatric and adult patients with bacterial meningitis, demonstrated **decreased overall hearing loss and neurologic sequalae for all organisms** when dexamethasone was administered.[8] **Reduced mortality** was noted in patients with *S. pneumoniae* meningitis.[8]

Appendix C: Indications for HSV evaluation and empiric treatment in children >4 weeks of age

- **Infants** aged 29-60 days of age: presenting with fever should be evaluated for neonatal HSV if there is maternal history of genital HSV *or* the infant presents with ≥1 of the following features:
    - Presence of vesicle(s)
    - Seizure(s)
    - Hypothermia
    - Mucous membrane ulcer(s)
    - CSF pleocytosis in the absence of positive Gram stain result
    - Leukopenia
    - Thrombocytopenia
    - Elevated LFTs.
- **Children >60 days of age**: presence of vesicular rash, personality changes, focal neurological findings, seizures, or imaging or EEG suggestive of temporal lobe involvement

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